On typical configurations the flow truncation is controlled at any coupling; the whole obstruction is the large-field tail

The condition $t\|G\|_\infty \lesssim 1$ of the Jacobian note can be evaluated rather than assumed. In the free theory the flowed magnetic field has

$$\langle |b_t|^2 \rangle = \frac{g^2}{16\pi^2 t^2}, \qquad \text{so} \qquad t\,\langle |b_t|^2 \rangle^{1/2} = \frac{g}{4\pi},$$

a pure number depending on the coupling alone and on no scale. Since the truncation error at range $R = \kappa\sqrt{8t}$ is $\exp[2t\|G\|_\infty - 2\kappa^2]$, a **typical** configuration is handled at any coupling by choosing

$$\kappa \;\gtrsim\; \sqrt{g/(4\pi)},$$

that is, a truncation range longer than the flow radius by the square root of the coupling. The obstruction is therefore not the typical field strength but the **tail**: $\|G\|_\infty$ is governed by the largest fluctuation anywhere in the volume, and for a Gaussian field it exceeds η/t with probability of order $e^{-c\eta^2/g^2}$ per correlation volume, small at weak coupling and only polynomially controlled by the volume. In the Euclidean framework that probability is a statement about the measure and can be absorbed, which is precisely the large-field estimate of a constructive renormalization group. In the Hamiltonian framework there is no measure to integrate over, only a state, and the same regions must be excluded by an operator statement. This locates the remaining difficulty in one place: **the large-field tail, and the absence of a Hamiltonian substitute for the probabilistic estimate that controls it.** Free-theory values are exact; their use for the interacting theory at scale ℓ is the standard perturbative estimate and is labelled as such. Constants explicit; nothing promoted.

> **Corrected in part.** The statements below are continuum statements at fixed physical scale. Within one lattice renormalization step the compactness of the gauge group bounds $a^2\|G\|_\infty$ by π and the flow time of a doubling is $a^2/2$, so the exponential factor is a pure number. A Jacobian locality estimate does not itself prove a relative Hamiltonian truncation bound; see the lattice-truncation note.

1. The flowed field strength in the free theory

From the free-field calculation §5c, the flowed magnetic two-point function at coincident points is

$$\langle b_{t,i}(0) b_{t,i}(0) \rangle = \int \frac{d^3k}{(2\pi)^3} \sum_i \tilde{G}_{ii}(k) = \int \frac{d^3k}{(2\pi)^3} \frac{g^2 k}{2}\, e^{-2tk^2} \cdot 2 = \frac{g^2}{2\pi^2} \int_0^\infty k^3 e^{-2tk^2}\, dk,$$

and $\int_0^\infty k^3 e^{-\alpha k^2} dk = 1/(2\alpha^2)$ with $\alpha = 2t$ gives

$$\langle |b_t|^2 \rangle = \frac{g^2}{2\pi^2} \cdot \frac{1}{8t^2} = \frac{g^2}{16\pi^2 t^2}, \qquad \langle |b_t|^2 \rangle^{1/2} = \frac{g}{4\pi\, t}.$$

The flow time is the only scale, so the dimensionless combination is

$$t\,\langle |b_t|^2 \rangle^{1/2} = \frac{g}{4\pi},$$

independent of t and of the lattice spacing. In the interacting theory at scale $\ell = \sqrt{8t}$ the same estimate with the running coupling $g(\ell)$ is the leading perturbative statement.

2. Typical configurations need only a longer range

Corollary 3 of the Jacobian note gives truncation error $\exp[2t\|G\|_\infty - 2\kappa^2]$ at range $R = \kappa\sqrt{8t}$. Replacing $\|G\|_\infty$ by the typical value of Section 1,

$$\varepsilon_{\text{typ}} \;\sim\; \exp\Big[\frac{g}{2\pi} - 2\kappa^2\Big],$$

so $\varepsilon_{\text{typ}} \le e^{-1}$ as soon as

$$\kappa^2 \;\ge\; \frac{g}{4\pi} + \frac{1}{2}, \qquad \text{i.e.} \qquad R \;\ge\; \sqrt{8t}\,\sqrt{\tfrac{g}{4\pi} + \tfrac{1}{2}}.$$

At $g = 1$ this is $R \approx 0.8\sqrt{8t}$; at $g = 4\pi$ it is $R \approx 1.2\sqrt{8t}$; the range grows only as $\sqrt{g}$. **On typical configurations the flow truncation is therefore controlled at every coupling**, at the price of an effective Hamiltonian whose range exceeds the scale by a factor $\sqrt{g/4\pi + 1/2}$.

This removes the reading of the small-field condition as a weak-coupling restriction. The condition $t\|G\|_\infty \lesssim 1$ is restrictive because of the supremum, not because of the typical size.

3. The tail is the obstruction

$\|G\|_\infty$ is the largest value of the field strength anywhere in the volume. For a Gaussian field of variance $\sigma^2 = g^2/(16\pi^2 t^2)$ the probability that $|b_t|$ exceeds η/t at a given point is

$$\mathbb{P}\Big(|b_t| > \frac{\eta}{t}\Big) \;\sim\; \exp\Big[-\frac{\eta^2}{2\sigma^2 t^2}\Big] = \exp\Big[-\frac{8\pi^2\eta^2}{g^2}\Big],$$

and the number of independent points at resolution $\sqrt{8t}$ in a volume V is $V/(8t)^{3/2}$, so

$$\mathbb{P}\Big(\|b_t\|_\infty > \frac{\eta}{t}\Big) \;\lesssim\; \frac{V}{(8t)^{3/2}}\,\exp\Big[-\frac{8\pi^2\eta^2}{g^2}\Big].$$

Two readings.

Euclidean. This is a statement about the Wilson measure, and the exponential beats the volume factor as long as $\eta^2 \gtrsim (g^2/8\pi^2)\log(V/(8t)^{3/2})$. Regions where it fails are the large-field regions, they occupy a fraction of the volume that is exponentially small in $1/g^2$, and a construction handles them by a separate argument whose cost is paid against that small probability. This is the structure of Balaban's programme (Commun. Math. Phys. 122 (1989) 355 and the accompanying series; metadata level) and the derivation above says why the split is forced: the truncation error is exponential in the local field strength, so the configurations must be sorted by it.

Hamiltonian. Here the object is a state, and $\|G\|_\infty$ enters an operator bound. There is no integration over configurations in which a rare region can be given small weight; the exponential factor $e^{2t\|G\|_\infty}$ multiplies an operator norm and must be controlled uniformly, or the argument must be localized so that a bad region affects only nearby terms. Localization of that kind is what the Lieb–Robinson bound of the LR note supplies in principle, and turning it into a substitute for the probabilistic estimate is the question this note isolates.

4. Consequences for the route

The window between the two methods is bounded and explicit. The cluster criterion of the T2 note holds for $g \ge g_0 = (48N^2/[(N^2-1)\beta_*])^{1/4}$; the flow truncation on typical configurations holds at every coupling with range factor $\sqrt{g/4\pi + 1/2}$, and the tail estimate degrades as $e^{-8\pi^2\eta^2/g^2}$. So the two do not fail in complementary regimes of the coupling, as an earlier reading suggested: the flow step is limited by the **tail at any coupling**, and the cluster step by the coupling itself. A proof must therefore control the large-field tail at every scale

where the cluster criterion has not yet taken over, which by the blocking note §3 is a number of steps logarithmic in the scale ratio, of order thirty for $SU(2)$ from a bare coupling $g^2_{\mathrm{UV}} = 1/2$.

The Euclidean framework is the right one for those steps. The suppression that makes the large-field regions harmless is a property of the measure, and it has no Hamiltonian counterpart. This is a reason, derived rather than conventional, why the constructive programme is carried out in the Euclidean formulation, and why the Hamiltonian route of this programme has reached its natural limit at exactly this point.

5. Consequence for STATE

The small-field condition is not a weak-coupling restriction: typical configurations are handled at any coupling by lengthening the truncation range as $\sqrt{g}$. The obstruction is the supremum over the volume, and its control is probabilistic, hence Euclidean. The Hamiltonian route therefore terminates here in a well-defined way, and the two questions it leaves are: whether a Lieb–Robinson localization can replace the large-field probability estimate, and, on the Euclidean side, whether the flow-truncation formulation of the step simplifies the existing constructive treatment. Either is a substantial project; the first is the one native to this programme.